

Automated Medicinal Plant Recognition using Capsule Networks

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Abstract- Medicinal plants are necessary for traditional as well as modern medicine, offering natural medication for a wide range of conditions. Accurate classification of these plants is essential to ensure their correct identification, which is crucial for safe usage, drug development, and conservation efforts. Misidentification can lead to ineffective treatment or even harmful effects. Additionally, proper classification aids in preserving biodiversity, supporting sustainable use, and protecting endangered species. The proposed system offers an innovative approach to medicinal plant identification by integrating Capsule Networks and the YOLO algorithm, addressing the critical need for accuracy in traditional medicine, conservation, and pharmacology. Manual identification is labor-intensive and error-prone, driving the development of automated systems. Conventional CNNs, while effective in classification, struggle with generalization and feature learning, and YOLO's use of rectangular bounding boxes can limit its accuracy in capturing complex leaf shapes. Capsule Networks maintain spatial hierarchies and handle affine transformations, enhancing feature extraction, while YOLO ensures real-time detection and accurate localization. The system was trained on a massive dataset of medicinal plant pictures, using upscaling and normalization techniques to improve precision. By leveraging the strengths of both Capsule Networks and YOLO, the architecture achieves superior performance, making it suitable for practical applications, including mobile devices.

Keywords— Capsule Networks, Machine Learning, Medicinal Plant, YOLO

I. INTRODUCTION

Over the past two years or so, ML techniques have indeed changed the way farming conducts plant identification and tracking. Actually useful in crop management, weed control, as well as precision agriculture, indeed accurate and automated detection would really be needed to optimize yield and resource use through timely and precise identification of plant species. Included in these are Conventional Neural Networks, or CNN, which are the deep learning models capable of learning hierarchical features that are widely used in plant detection tasks. Still, it is common that these models cannot precisely represent spatial relationships between plants as well as creating differences in their appearance due to its common issue on farming wherein plants can have different orientations under different lighting conditions. Besides that, CNNs face various overfitting problems and entail huge labeled datasets that degrade their performance in real-world tasks. For this reason, Geoffrey Hinton et al introduced a new structure called Capsule Networks-CapsNets to overcome the weaknesses involved with traditional CNN models. Unlike CNNs, CapsNets are actually well-crafted to better

capture spatial hierarchies through the use of capsules, groups of neurons that latch onto inherent object properties like pose and rotation. For the learning process about complex and variable shapes in plant detection, as applied in the Capsule Network, it retains spatial relationships between different features of an object. On application of Capsule Network on automated plant detection, showing how the architecture it gives rise to could make much more accurate or robust than current methods. Applying CapsNet enabling understanding and generalizing over this diversity of appearances of plant in a way that highly contributes to making more efficient and scalable solutions in agricultural technology.

II. RELATED WORKS

S. S. Anvitha et al., has concentrated upon developing an image classifier which classifies different fine grained floral images based upon their species. Classification of flowers has many advantages in various fields like Science, Medical, Industries, etc. Through various Convolutional neural network models the work is able to propose an efficient model based on fine-tuned MobileNetV3 [1] architecture and self-attention mechanisms that classifies fine-grained flower images into their respective species.

A. Lakshmanarao et al., have developed a comprehensive method for recognising of medicinal plant species depending on the DL features generated using VGG16 architecture, these features were augmented with traditional methods to develop the final feature vector. Later, the ML models, that is, SVM [2], Random Forests, Decision Trees, logistic regression and KNN were applied to the final dataset in order to classify the medicinal plant species. Using comparative analysis, the model that performed the best with the highest accuracy recorded was random forest at 98.3%. It highlighted the use of integrated deep methods in more accurately detecting plants.

N. Naik et al., an innovative Android application is designed to identify several herbal plant species just by scanning their leaves. Using Indian herbal plant species dataset, the application combines the use of image processing techniques for accurate leaf detection, feature extraction, training the model and then finally apply classification [3] techniques to match the leaf accurately to its appropriate species. Effective features such as area, length, and width of the leaf also contribute to the model's excellent training.

S. S. Srmbickal et al., proposed a method that uses deep learning to identify the given plant using the leaves of the

plant by using a pretrained Convolutional Neural Network (CNN) model namely ResNet50 with MobileNet [4] for the extraction of latent vector which will then be passed to K-Nearest Neighbour (KNN) algorithm in order to identify the plant class, the best accuracy of the proposed models comes out to be 97.65%.

M. Sangeetha et al., provides a new CNN-based approach to predict medicinal leaf types. A CNN [5] model was carefully trained over this dataset. This sophisticated dataset helps the CNN model to learn about various varieties of medicinal leaves.

J. Sinha et al., proposed a project for the study of an Ayurvedic leaf by the application of Transfer Learning based Convolution Neural Networks. The accuracy of Herb identification has reached a peak of 97.97% in testing accuracy for DenseNet121 Convolution Neural Network. InceptionV3, MobileNetV2 [6] and VGG16 models provides a test accuracy of 97.59%, 96.21% and 93.79% independently.

Chalmers J et al., emphasizes the vision-based system in the recognition of herbal leaf by developing high-level sophisticated deep learning models like YOLOV5, YOLOV8 [7], and Resnet-50. The evaluation metric is presented to validate that YOLOV8 obtained 95.2% accuracy.

R. Nidhya et al., suggested a new way for the classification and detection of medicinal plant species using machine learning approaches, particularly Random Forest [8]. Classification method applied in this case is Random Forest. A very impressive result with 98% classification accuracy is achieved.

R. Thorat et al., assist in managing the problems of these affliction and prevail over the disadvantage of the uninterrupted human monitoring. In this article, the authors have illustrated the difference in the performance of a Convolutional neural network (CNN) with several other architectures such as DenseNet, MobileNet, ResNet [9]. Convolutional neural network was used for plain images of tomato leaves with classification of several tomato leaf diseases. After selecting CNN, it was possible to reach accuracy in classification equal to 95.8%.

R. G. Tiwari et al., proposed a new feature-fusion network called "MedLeafNet" and classification of Indian medicinal leaves is presented herein. Best features of Convolutional Neural Network (CNN) modules is linked by MedLeafNet [10] for the extraction of spatial features and Transformer modules for sequence modeling using hybrid architecture. For this paper, 80 different types of Indian leaves were taken that had well-known therapeutic properties. MedLeafNet achieved maximum accuracy as high as 98.97% when trained by Adam optimizer as demonstrated through extensive testing.

M. T. Islam et al., presents a robust, accurate and feasible system in the area of plant identification of medicinal plants from images captured through a smartphone in the field. In the proposed system, the pre-trained ResNet50[11] model is utilized for the extraction of features by a cascaded architecture and optimized for PSO in SVM for classification of the plant. It classifies seven medicinal

plants with 97.60% accuracy proposed by ResNet50-PSO-SVM network.

Y. Zhang et al., proposed a function called LReLU-Softplus. It is the combination of Leaky Rectified Linear Unit and Softplus function. For the selfmade women's clothing image database, LReLU-Softplus [12] achieved recognition with 97.4%, while the performance of Softplus is 96.3%, and that of ReLU at 93.2% within the range of the iteration cycles of the study. The best recognition rate reached in the Mnist digit database is LReLU-Softplus, at 98.4%, followed by that of Tanh at 97.7% and then Sigmoid's rate at 97.6%.

P. Salian et al., have made it possible to automate the identification of medicinal plants from photographs of their leaves. The VGG16 architecture based deep Learning system built at NPI&O is also professionally well known as MPInet[13] which identifies 10 medicinal plants that are mostly available in populous regions is trained on self collected data. There are a total of 3000 images representing ten classes present in the dataset. The constructed model produced the result of 96.4 percent in terms of the precision accuracy, 96.6 percent in terms of the precision value, recall value of 99.5% and an F1 Score of 96.8%. In mobile appliances MPInet is embedded for making it useful for learning purposes.

S. Pargaien et al., used three different models of machine learning which can be named as; Logistic Regression combined with Inception-v3 feature extractor [14] with Neural Network (NN) confusion matrix and Random Forest confusion matrix (the Orange Classification Tree) with Inception-v3 feature extractor. Average response for each model employed was prepared and the best one turned out to be 98.5%. For each of the employed model that was developed, ROC curve was plotted for fair comparison and representation purposes.

Y. Sharab et al., Propose CNN-based approach with deep learning by using VGG-16 model [15]. The capacity of CNN approach in learning to represent the complex objects in images, which identifies and classifies medicinal plants correctly. This method enables such a reliable tool given the correct classification of images through these different phases of development, lighting conditions, and imaging acquisition. We also could verify that deep learning techniques allow plants to be classified at an astonishing recognition rate of 96%.

III. PROPOSED METHODOLOGY

It is an automatically proposed system of medicinal plant recognition with high involvement and advanced utilization of enhanced capabilities to achieve better accuracy and efficiency in classification. Recognition logically consisted of distinct, well-connected modules that were both important and successive in the process of recognition. Figure 1 shows the architecture of this objective, data were collected in the form of very high resolution pictures of medicinal plants which was further augmented and normalized to robust to the model. The training and the testing sets of the data ascertain an eligible testing course.

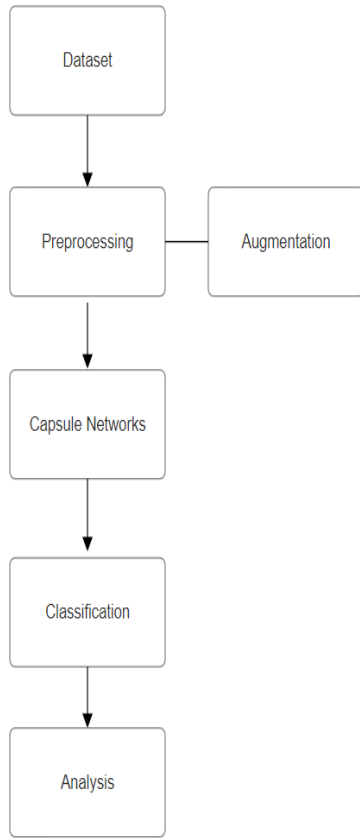


Fig.1. Proposed System

Modules:

A. Dataset

The dataset is obtained from Kaggle as Medicinal Leaf Dataset which is shown in Figure 2. The dataset consist of 1500 images of forty species. The dataset is split into three sub-sets: testing, training and validation in a proportion of sizes 0.7:0.2:0.1, that is, training, testing and validation subset proportions, respectively.



Fig.2. Dataset Image

B. Data Preprocessing

Data Preprocessing is in fact the central module of this proposed system wherein raw data in the form of images is prepared. Samples of images of different medicinal plants are taken so that the dataset would contain images of various species. Augmentation techniques such as scaling,

flipping and rotation is used to artificially increase variability in the dataset. Normalization will be applied to the dataset so that pixel values in all the images are standardized, and, hence, the model will actually learn better. The dataset is segregated as a training set and a test set usually in an 80/20 ratio so that the model may be tested on unseen data, therefore giving proper measure of its performance.

C. Feature Extraction

In the module of Feature Extraction, Capsule Networks applies its strength core in analyzing the processed images. Its work on the spatial hierarchies and relationships that aim at preserving the image data makes it more flexible to understand plant morphology than through a traditional convolutional network. In fact, a capsule at every stage of the network will try to capture certain features within the images taken- its size, orientation, and even position like leaves and flowers. This allows for the learning of model features describing another species of the plant, independent of differences in viewpoints or environmental conditions and maps such features into a high-dimensional space for improvement in classification performance.

D. Model Training

Model Training aims at optimizing the classification for the Capsule Network. Images and labels are passed through it during the train phase. It faces a series of iterates in the training phase and adjusts its weights and biases with an algorithm of backpropagation in order to minimize the error in the model's classification. The main hyper-parameters that were varied during this training phase in order to enhance convergence as well as the overall performance of the model are learning rate and batch size. Other techniques like dropout regularization may be used in an attempt to overcome overfitting. Then the overall model will generalize very well to unseen and new data. Training has been accomplished until sufficient performance of the model on a training set has been achieved where the loss decreased and accuracy increased.

E. Model Evaluation

At the end of the model, the Model Evaluation module tests its performance on the set-aside testing dataset. This proves that the model can classify the sample correctly and correct identification of medicinal plants. Metrics known as F1-score, recall, accuracy and precision for a full unveiling of what are the pluses and minuses of the model. Accuracy checks if the majority of classifications are correct, while precision and recall give a good idea of how well the model performs at some classes, how well it finds true positives or minimizes false positives. The harmonic mean F1-score combines precision and recall for an ideally balanced measure of models' performance. Careful testing to ensure the model is well-designed towards real-world implementation settings.

1) CAPSULE NEURAL NETWORK

A group of neurons called capsule where the instantiation parameters of a given type of an entity is represented by activity vector. In particular, capsules encapsulate all meaningful information about the state of the feature being detected in vector form. Holding spatial correlation between data features is variable in the ability of CapsNet. The capsule is a vector whose length determines the

probability of detection of a feature. Table.1 shows the difference between conventional neuron and capsule.

TABLE I. TRADITIONAL NEURON VS CAPSULE

Traditional Neuron vs Capsule			
Input from low level capsule/neuron		Vector (u_i)	Scalar (x_i)
Operation	Weighting	$\hat{u}_{j i} = W_{ij}u_i$	-
	Sum	$s_j = \sum c_{ij} \hat{u}_{j i}$	$a_j = \sum_i w_i x_i + b$
	Nonlinear activation	$s_j = \frac{\ s_j\ ^2}{1 + \ s_j\ ^2} \frac{s_j}{\ s_j\ }$	$h_j = f(a_j)$
Output		Vector (V_j)	Scalar (h_j)

As follows, the operations that capsules carry:

1. Matrices Input Vectors Multiplication
2. Vector-to-vector nonlinearity
3. Weighted sum of the input vectors
4. Scalar Weighting of Input Vectors

Let us consider a vector $u = (u_1; u_2; \dots; u_n)$ to use as input to the current capsule. There is a matrix W which encodes an important relationship between low-level features and high-level feature. The selection among capsules of higher levels is done through the dynamic routing algorithm which corrects the inner matrix C .

CapsNet Architecture

The two major parts of the CapsNet include encoder and decoder

Encoder:

It accepts 28x28 and runs the routing algorithm. Output of this encoder is 10-dimensional vectors of the sizes of DigitCaps'.

Decoder

It collects the correct output from DigitCaps and attempts to reconstruct it. The first three layers are an encoder and the subsequent three layers are a decoder through the regularization method.

Convolutional layer

Detection of basic features: the first layer. There are 256 kernels with stride 1 of size 9x9x1 and activation function as ReLU .

PrimaryCaps layer

The 32 low-level capsules are produced in this layer with 9x9x256 convolution. kernels with stride 2). The aggregation of the elementary features into capsules represents the core problem.

DigitCaps layer

The following layer performs dynamic routing with 10-digit capsules.

Fully connected layer

This constructs a layer wise decoder with classification.

IV. RESULTS AND DISCUSSION

For every predictive model 1,024 sequences per epoch, and did 100 iterations in mini-batch mode. We split our data into independent pieces that we train each model 10 times. Then we report mean evaluation metrics over 10 testing results. The training and validation accuracy is shown in Figure 3. The loss curve is shown in Figure 4.

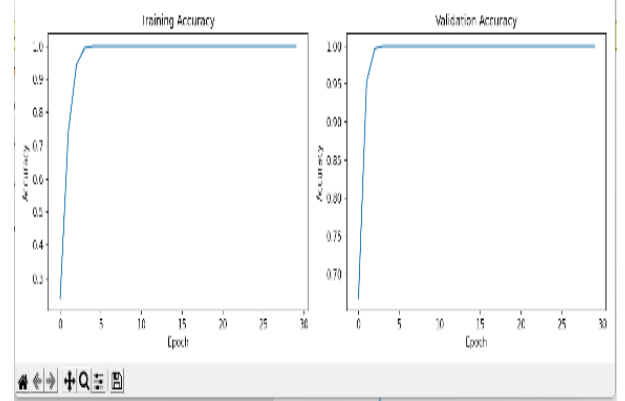


Fig.3 Accuracy Curve

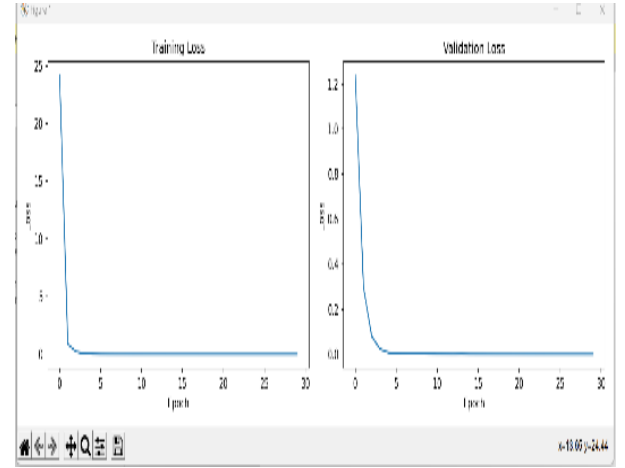


Fig.4 Loss Curve

Accuracy of validation as well as loss curves were quite stable and converging; excellent generalization to unseen data is represented. It is very vital, especially in the case of medical imaging, mainly because the cases that the model will make in cardiomegaly are very critical in clinical decisions. Trends in curves indicate reliance and efficiency of the proposed deep learning approach. The input image which is passed into the system is classified according to the model trained and the output is shown in Figure 5.





Fig. 5. Classification results

TABLE II. PERFORMANCE ANALYSIS

Models	Precision	Recall	Accuracy	F1-Score
GoogleNet	86.49	79.42	86.19	90.29
InceptionNet	91.31	86.318	90.75	93.08
ResNet	91.28	80.08	85.4	91.71
CapsNet	97.71	94.71	94.99	98.6

Table.2 shows Performance Analysis of F1 Score, Precision, Accuracy and Specificity of the suggested method with several Models.

As such, the Proposed Model indicates excellence in words of studied models in terms of performance metrics. Such F1-score, recall, precision, and accuracy produced by this model is simply remarkable. With this model, impressive proficiency in recognizing true positives about cardiomegaly with minimal false positives, exhibiting 97.76% precision is given. That is simply saying that the model is very good at the recognition of actual cases of the leaves.

More to that, the Proposed Model has an exceptionally good recall of 94.78 percent which indicates that it captures large percentage of the true positive cases existing in the dataset. Such a recall indicates that large proportion of cases of cardiomegaly that exist are almost certainly captured within the images.

With a report of the accuracy level of the proposed model at 95.1%, an overall correctness in classifying both the positive and negative instances was proven. This really means that the model is very reliable and effective in differentiating all instances from cases in the dataset.

Lastly, the F1 Score of the Proposed Model is excellent and stands at 98.7%. F1Score has properties of both precision and recall to completely evaluate the model. With high chances of false positives, and not misclassifying falsely in cases of a leaf, the former describes robustness in all aspects of the model.

V. CONCLUSION

The proposed work demonstrated the efficient applicability of Capsule Networks toward unsupervised medicinal plant recognition. While the method uses rigorously annotated practices and high-resolution images, it accurately captures

spatial hierarchies and the complex relationships involved in the morphological characteristics of the plants while achieving correct classification of targeted species. The F1-score, precision, recall, and accuracy score performance result depict most important points in proving that it holds the capacity to become a useful tool for reliable identification of plants. Beyond the effects of simple classification, our findings may serve as an important source for herbalists, researchers, and even conservationists who might help preserve and present traditional medicinal knowledge. This can then be integrated into mobile applications which would then use to identify medicinal plants at a real-time basis that would, in turn, serve to improve and enhance the sustainable use of local flora. Thus, further work includes making the model even more robust with its addition of diverse datasets as well as further investigation of other deep learning techniques that could improve the accuracy of classification. These would fine-tune our approach toward more effective biodiversity conservation and the sustainable utilization of medicinal plants for further engendering much more appreciation of these priceless resources that nature has to offer.

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